

Radiation and Variable Viscosity Effects on Hydromagnetic Flow Along a Moving Vertical Plate Embedded in a Porous Medium

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Abstract: This paper investigates the effects of radiation and variable viscosity on a viscous incompressible and electrically conducting fluid over a continuously moving vertical infinite plate with uniform suction and heat flux, embedded in a porous medium. The governing partial differential equations are transformed into ordinary differential equations by using similarity variables and then the resultant equations are solved numerically by employing the fourth order Runge-Kutta method along with shooting technique. The effects of various governing thermophysical parameters on the velocity, temperature and skin-friction coefficient are shown in figures and tables and analyzed in detail.

Keywords: MHD, Variable Viscosity, Radiation, Porous medium and Vertical plate

1. INTRODUCTION

Flow and heat transfer in the boundary layer induced by a moving surface in a quiescent fluid are important in many engineering applications, for examples, in the extrusion of polymer sheet from a die, the cooling of an infinite metallic plate in a cooling bath, glass blowing continuous casting and spinning of fibers. Sakiadis [1] studied the boundary layer flow over a continuous solid surface moving with constant velocity in an ambient fluid. The flow is quite different from the boundary layer flow over a semi-infinite flat plate due to the entrainment of the ambient fluid. Crane [2] studied the boundary layer flow caused by a stretching sheet whose velocity varies linearly with the distance from a fixed point on the surface.

The problem of free convection under the influence of a magnetic field has attracted the interest of many researchers in view of its applications in geophysics and astrophysics. Engineers employ principles of magnetohydrodynamics (MHD) in the design of heat exchangers, pumps, in space vehicle propulsion, thermal protection, control and re-entry and in creating novel power generating systems. Vajravelu and Hadjimecoulou [3] studied the flow and heat transfer characteristics in an electrically conducting fluid near an isothermal stretching sheet. Soundalgekar *et al* [4] analyzed the problem of free convection effects on Stokes problem for a vertical plate under the action of transversely applied magnetic field. Zucco [5] analyzed the hydromagnetic convection past a flat plate.